Volatilni cast:

```
F_i,_t=\sum_c\omega_i,_c[w_g\tau(realGDPgrowth_c,_t)+w_y\tau(realIncomeGrowth_c,_t)-w_u\tau((currentUnemployment-normalUnemployment)_c,_t)+w_p\tau(PMI_c,_t-50)+w_\pi*I_\pi,_c,_t]
```

Copy

$$F_{i,t} = \sum_c \omega_{i,c} [w_g \tau(realGDPgrowth_{c,t}) + w_y \tau(realIncomeGrowth_{c,t}) - w_u \tau((currentUnei$$


```
F_i,_t=\sum_c\omega_i,_c[w_g\tau(realGDPgrowth_c,_t)+w_y\tau(realIncomeGrowth_c,_t)-w_u\tau((currentUnemployment-normalUnemployment)_c,_t)+w_p\tau(PMI_c,_t-50)+w_\pi*I_\pi,_c,_t]
```

Copy

$$\tau((currentUnemployment - normalUnemployment)_{c,t}) + w_p \tau(PMI_{c,t} - 50) + w_\pi * I_{\pi,c,t}]$$

F_i,_t = stav svetove ekonomiky ted (z “pohledu” komodity)

\sum_c = suma ze vsech statu

\omega_i,_c = dulezitost statu (znak “c”) pro komoditu (znak “i”)

PMI_c,_t – 50 = aby ta 50 byla jako prostredek normy (teda 50 = normal, <50 = spad, >50 = rust)

w_g = vaha rustu HDP

w_y = vaha rustu realnych prijmu

w_u = vaha nezamestnanosti

w_p = vaha PMI

w_pi = vaha inflacni stability

Inflacni cast (doplnek k volatilni casti):

```
I_\pi,_c,_t=2\exp[-(\frac{\pi_c,_t-\pi^*_c}{\sigma_\pi,_c})^2]-1
```

Copy

$$I_{\pi,c,t} = 2\exp[-(\frac{\pi_{c,t} - \pi^*_c}{\sigma_{\pi,c}})^2] - 1$$

I_\pi,_c,_t = jak dobre nebo spatne inflace ovlivnuje ekonomiku

\pi_c,_t = realni inflace

\pi_c,_t = normalni inflace

\sigma_\pi,_c = prijatelna odchylka od normy

Zakladni cast:

```
L_i,_t=\sum_c\omega_i,_c[v_q\tau(GDP^{PPP}_{c,t})+v_y\tau(real
Income_c,_t)+v_f\tau(FDI/GDP_c,_t)+v_nT(populationGrowth_c,_t)]
```

Copy

$$L_{i,t} = \sum_{c} \omega_{i,c} \left[v_q \tau(GDP_{c,t}^{PPP}) + v_y \tau(realIncome_{c,t}) + v_f \tau(FDI/GDP_{c,t}) + v_n T(populatio \right.$$


```
L_i,_t=\sum_c\omega_i,_c[v_q\tau(GDP^{PPP}_{c,t})+v_y\tau(real
Income_c,_t)+v_f\tau(FDI/GDP_c,_t)+v_nT(populationGrowth_c,_t)]
```

Copy

$$_c \left[v_q \tau(GDP_{c,t}^{PPP}) + v_y \tau(realIncome_{c,t}) + v_f \tau(FDI/GDP_{c,t}) + v_n T(populationGrowth_{c,t}) \right]$$

L_i,_t = schopnost svetove ekonomiky dlouhodobe mit potrebu v presne komodite

GDP^{PPP}_{c,t} = proste HDP po PPP konkretniho statu

FDI/GDP_c,_t = FDI presne jako cast HDP

populationGrowth_c,_t = tempo rustu poctu obyvatel (ne samotny pocet)

v_q = vaha HDP po PPP

v_y = vaha prijmu obyvatel

v_f = vaha primych investici ze zahranici

v_n = vaha poctu obyvatel

“Distribucni” cast:

```
\lambda_A,_i(\tau)=\lambda^\infty_{A,i}+(\lambda^0_{A,i}-
\lambda^\infty_{A,i})\exp(-\frac{\tau}{\vartheta_{A,i,z_t}})
```

Copy

$$\lambda_{A,i}(\tau) = \lambda^{\infty}_{A,i} + (\lambda^{0}_{A,i} - \lambda^{\infty}_{A,i}) \exp(-\frac{\tau}{\vartheta_{A,i,z_t}})$$

\lambda^0_{A,i} = vliv volatilni casti

\lambda^\infty_{A,i} = zbyvajici vliv volatilni casti na dlouhe distance

\vartheta_{A,i,z_t} = rychlost prechodu od “volatilniho stavu” k “normalnimu zakladnimu stavu” ekonomiky

z_t = “kvalita” stavu trhu

Hlavni formula indexu:

```
A^{(\ell)}_{i,t}=\frac{1}{2}[1+e^{-\kappa_{A,i}|F_i,_t-L_i,_t|}\tanh(\beta_A,_i[\lambda_A,_i(\tau_\ell)F_i,_t+(1-\lambda_A,_i(\tau_\ell))L_i,_t+\chi_iF_i,_tL_i,_t\frac{F_i,_t+L_i,_t}{2}])]
```

Copy

$$A_{i,t}^{(\ell)} = \frac{1}{2}[1 + e^{-\kappa_{A,i}|F_{i,t}-L_{i,t}|} \tanh(\beta_{A,i} [\lambda_{A,i} (\tau_\ell)F_{i,t} +(1 - \lambda_{A,i} (\tau_\ell))L_{i,t} +\chi_i F_{i,t} L_{i,t} \frac{F_{i,t} +L_{i,t}$$

```
A^{(\ell)}_{i,t}=\frac{1}{2}[1+e^{-\kappa_{A,i}|F_i,_t-L_i,_t|}\tanh(\beta_A,_i[\lambda_A,_i(\tau_\ell)F_i,_t+(1-\lambda_A,_i(\tau_\ell))L_i,_t+\chi_iF_i,_tL_i,_t\frac{F_i,_t+L_i,_t}{2}])]
```

Copy

$$) = \frac{1}{2}[1 + e^{-\kappa_{A,i}|F_{i,t}-L_{i,t}|} \tanh(\beta_{A,i} [\lambda_{A,i} (\tau_\ell)F_{i,t} +(1 - \lambda_{A,i} (\tau_\ell))L_{i,t} +\chi_i F_{i,t} L_{i,t} \frac{F_{i,t} +L_{i,t}}{2}])]$$

A^{(\ell)}_{i,t} = index ekonomicke activity komodity “i” primo ted “t” v jedne z budoucich casovych pozici “\ell”

F_i,_t = volatilni cast

L_i,_t = zakladni cast

\lambda_A,_i(\tau_\ell) = distribuce vlivu mezi rychlou a zakladni casti

\chi_i = zesileni efektu spoceneho rustu nebo spadu (volatilni a zakladni)

e^-{\kappa_{A,i}|F_i,_t-L_i,_t| = oslabnuti efektu pri rozporu

\kappa_{A,i} = sila oslabnuti efektu

\beta_A,_i = citlivost indexu ke zmenam ekonomiky